compute reachability distances only once, and try several values for ε (up to εMAX) in this second step, with minimal computational effort.

Listing 12.9 OPTICS opticsCluster

```
function opticsCluster(ordering, reachabilityDistances, eps)
         #1
  currentClusterIndex ← 0
                           #2
  incrementCurrentIndex ←
false                                          #3
  clusterIndices ← -1 (" p ∈
points)                                      #4
  for i in {0, .. |ordering|} do
                  #5
    if reachabilityDistances[ordering[i]] == null or
         reachabilityDistances[ordering[i]] > eps then
              #6
      incrementCurrentIndex ←
true                                        #7
    else
      if incrementClusterIndex
then                                      #8
        currentClusterIndex ← currentClusterIndex +
1                #9
        clusterIndices[ordering[i-1]] ←
currentClusterIndex          #10
      clusterIndices[ordering[i]] ←
currentClusterIndex              #11
      incrementCurrentIndex ←
false                                        #12
  return clusterIndices
                                   #13
```